

CCNA 200-301 Notes

Chapter 08 - First Network Lab (Packet Tracer)

This chapter covers the first practical lab using Cisco Packet Tracer.

Objectives

- Build a LAN with one switch and three PCs.
- Configure IPv4 manually.
- Test connectivity using ipconfig and ping.
- Troubleshoot common network problems.

Topology

PC0: 192.168.1.1

PC1: 192.168.1.2

PC2: 192.168.1.3

Subnet Mask: 255.255.255.0

Commands

ipconfig

ipconfig /all

ping 192.168.1.2

ping google.com

ping -l 200 google.com

ping -t google.com

Troubleshooting

If Request Timed Out appears, check the target device, cable, IP settings, subnet mask, firewall and ensure devices are in the same network.

ARP Note

The first ping triggers an ARP Request (Broadcast). After MAC learning, ICMP packets are sent as Unicast.

Exam Tips

Draw the topology first, verify settings with ipconfig/ipconfig /all, then test with ping.